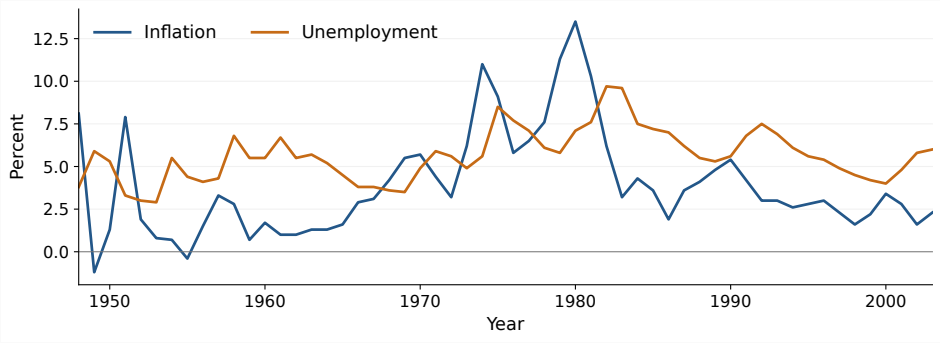
# Lecture 7 — Introduction to Time Series

Introductory Econometrics — Télécom Paris

Màxim Sandiumenge i Boy

Reading: Wooldridge (5th ed.), selected sections of chapters 10–11

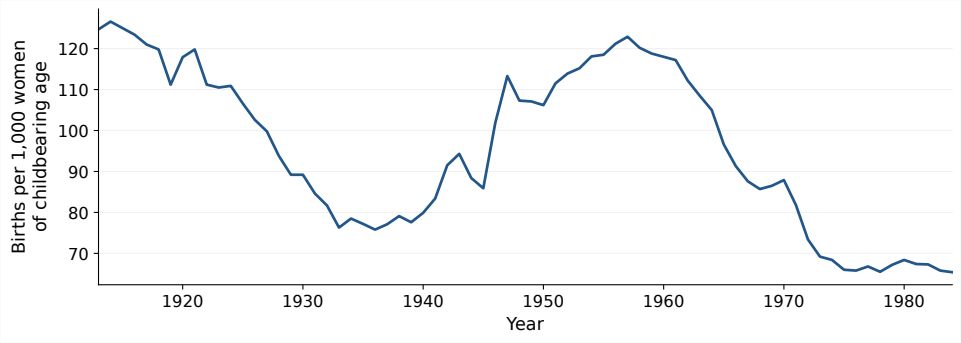
# Inflation and unemployment over time




US annual data, 1948–2003. Source: Wooldridge, PHILLIPS.

Do inflation and unemployment move together? Their timing and persistence matter when we use regression to study their relationship.

# Fertility over time




US annual data, 1913–1984. Source: Wooldridge, `FERTIL3`.

Fertility changes substantially over decades. Can tax incentives help explain these changes, and might their effects take several years?

# Why time order changes the analysis

Temporal ordering is a defining characteristic of time series data.

In analyzing time series data, we recognize that the past can affect the future, but the future cannot affect the past.

The outcomes of these variables are not known in advance, so we view them as random variables.

# A stochastic process and its observed history

**Time series data as the outcome of random variables.**

A **stochastic process**, also called a **time series process**, is a collection of random variables $\{y_t\}$ indexed by time.

$$\underbrace{(y_1, \ldots, y_T)}_{\text{random variables}} \quad \text{and} \quad \underbrace{(y_1^{\text{obs}}, \ldots, y_T^{\text{obs}})}_{\text{one observed realization}} .$$

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A time series data set records one possible outcome, or realization, of the stochastic process.

The set of all possible realizations plays the role of the **population** in cross-sectional analysis.

The **sample size** $T$ is the number of time periods over which we observe the variables of interest.

# Lags, differences and growth rates

$$\begin{aligned}
\text{First lag:} \quad & y_{t-1}, \\
\text{First difference:} \quad & \Delta y_t = y_t - y_{t-1}, \\
\text{Approximate growth rate:} \quad & 100\Delta \log y_t \approx 100\frac{y_t - y_{t-1}}{y_{t-1}} \quad (y_t, y_{t-1} > 0).
\end{aligned}$$

The log approximation is most accurate for small proportional changes. If a price index rises from 100 to 103, inflation is 3%. If inflation then rises from 3% to 4%, its change is **1 percentage point**.

# Static Models

A static model is a time series model that can be easily estimated by OLS.

Consider two variables, $y$ and $z$, observed contemporaneously at dates $t = 1, \ldots, T$. A static model relating $y$ to $z$ is

$$y_t = \beta_0 + \beta_1 z_t + u_t.$$

We can use it to study the contemporaneous tradeoff between $y$ and $z$.

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**Example: a form of the Phillips curve.**

$$infl_t = \beta_0 + \beta_1 unem_t + u_t.$$

Here $infl_t$ is the annual inflation rate and $unem_t$ is the unemployment rate in the same year.

# Finite Distributed Lag Models

A finite distributed lag model is another time series model that can be easily estimated by OLS.

An economic response can also take time. Consider an outcome $y_t$ and a variable $z_t$ whose current and past values may affect it. A model of order $q$ is

$$y_t = \alpha + \delta_0 z_t + \delta_1 z_{t-1} + \cdots + \delta_q z_{t-q} + u_t.$$

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**Example: fertility and tax incentives.** A change in the tax incentive to have children may affect fertility over several years:

$$gfr_t = \alpha + \delta_0 pe_t + \delta_1 pe_{t-1} + \delta_2 pe_{t-2} + u_t.$$

Here *gfr* is births per 1,000 women of childbearing age and *pe* is the real personal tax exemption. This example has $q = 2$.

# Finite Distributed Lag Models: Responses over time

$y_t = \alpha + \delta_0 z_t + \delta_1 z_{t-1} + \delta_2 z_{t-2} + u_t$, with $z_s = c$ for all $s < t$. At $t$, increase $z$ by one unit. Hold disturbances at their pre-shock value.

| | Response: $y_{t'} - y_{t-1}$ | |
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| Date $t'$ | Increase only at $t$ | Permanent increase |

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| $t+1$ | $\delta_1$ | $\delta_0 + \delta_1$ |
| $t+2$ | $\delta_2$ | $\delta_0 + \delta_1 + \delta_2$ |

$\delta_0$: **impact propensity** or **multiplier**; $\delta_0 + \delta_1 + \delta_2$: **long-run multiplier** or **propensity (LRP)**.

Substantial correlation between lags of $z$ (multicollinearity) reduces the precision of $\hat{\delta}_j$.

# Finite Sample properties of OLS under Classical Assumptions

# Time-series regression and strict exogeneity

Write the population model as

$$y_t = \beta_0 + \beta_1 x_{t1} + \cdots + \beta_k x_{tk} + u_t, \qquad t = 1, \ldots, T.$$

Let $x_t = (x_{t1}, \ldots, x_{tk})'$ contain the explanatory variables. The rows of $X$ are $x_t'$.

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**TS.1 — Linear in Parameters.** The stochastic process $\{(x_{t1}, \ldots, x_{tk}, y_t) : t = 1, \ldots, T\}$ follows this linear model, where $\{u_t : t = 1, \ldots, T\}$ is the sequence of errors or disturbances.

**TS.2 — No Perfect Collinearity.** In the sample, no explanatory variable is constant or a perfect linear combination of the others.

# Zero Conditional Mean and strict exogeneity

**TS.3 — Zero Conditional Mean (strict exogeneity).** For each $t$, the expected error given the explanatory variables for *all* time periods is zero:

$$\mathbb{E}(u_t \mid X) = 0, \qquad t = 1, \ldots, T.$$

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- **Contemporaneous exogeneity**: $\mathbb{E}(u_t \mid x_t) = 0$ conditions only on current regressors; it is weaker than TS.3.
- In cross sections, we did not explicitly restrict $u_i$ relative to other observations' regressors $x_{-i}$: random sampling made them independent.
- Random sampling is generally inappropriate for time series, motivating the stronger assumption.
- TS.3 puts no restriction on correlation within $\{x_t\}$ or within $\{u_t\}$ across time.

# Why strict exogeneity can fail

In $y_t = \beta_0 + \beta_1 z_t + u_t$, TS.3 requires $\mathbb{E}(u_t \mid z_1, \ldots, z_T) = 0$.

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**Omitted variables.** An omitted determinant of $y_t$ belongs to $u_t$; correlation with a regressor at any date can violate TS.3.

**Measurement error in a regressor.** The observed regressor can share its measurement error with the regression disturbance, making them correlated.

**Feedback from $u_t$ to future $z$.** An unexpected change in $y_t$ can influence future $z$, correlating $u_t$ with future regressors.

For example, an unexpected rise in crime may lead to more police next year. TS.3 rules out this feedback from $y_t$ to future $z$ through $u_t$.

# Unbiasedness of OLS

**Theorem 10.1.** Under TS.1, TS.2 and TS.3, OLS is unbiased conditional on $X$, and therefore unconditionally:

$$\mathbb{E}(\hat{\beta}_j \mid X) = \beta_j, \qquad \mathbb{E}(\hat{\beta}_j) = \beta_j, \qquad j = 0, \ldots, k.$$

**Proof.** Let $\beta = (\beta_0, \ldots, \beta_k)'$. Under TS.1–TS.2,

$$\hat{\beta} = (X'X)^{-1}X'y = \beta + (X'X)^{-1}X'u.$$

Conditional on $X$, $(X'X)^{-1}X'$ is fixed. TS.3 gives $\mathbb{E}(u \mid X) = 0$, hence

$$\mathbb{E}(\hat{\beta} \mid X) = \beta + (X'X)^{-1}X'\mathbb{E}(u \mid X) = \beta.$$

Taking expectations gives $\mathbb{E}(\hat{\beta}) = \beta$.

# Gauss–Markov assumptions and OLS sampling variances

**TS.4 — Homoskedasticity.** Conditional on $X$, the error variance is the same at every date:

$$\mathrm{Var}(u_t \mid X) = \mathrm{Var}(u_t) = \sigma^2, \qquad t = 1, \ldots, T.$$

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**TS.5 — No Serial Correlation.** Conditional on $X$, errors at different dates are uncorrelated: $\mathrm{Corr}(u_t, u_s \mid X) = 0$ for $t \neq s$.

This restricts the errors, not temporal correlation in the explanatory variables. In cross sections, random sampling made errors from distinct observations independent.

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**Theorem 10.2 — OLS Sampling Variances.** Under the **Gauss–Markov Assumptions (TS.1–TS.5)**,

$$\mathrm{Var}(\hat{\beta} \mid X) = \sigma^2 (X'X)^{-1}.$$

# Estimating the error variance and the Gauss–Markov theorem

**Theorem 10.3 — Unbiased Estimation of $\sigma^2$.** Under TS.1–TS.5, the residual variance estimator is unbiased:

$$\hat{\sigma}^2 = \frac{\sum_{t=1}^{T} \hat{u}_t^2}{T-k-1}, \qquad \mathbb{E}(\hat{\sigma}^2 \mid X) = \sigma^2, \qquad \mathbb{E}(\hat{\sigma}^2) = \sigma^2.$$

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**Theorem 10.4 — Gauss–Markov Theorem.** Under TS.1–TS.5, OLS is the **best linear unbiased estimator (BLUE)** conditional on $X$.

For any other estimator $\tilde{\beta}$ linear in $y$ and unbiased conditional on $X$,

$$\mathrm{Var}(\tilde{\beta} \mid X) - \mathrm{Var}(\hat{\beta} \mid X) \succeq 0.$$

No linear combination of its coefficients has a smaller conditional variance than the corresponding OLS combination.

# Inference under the classical linear model

**TS.6 — Normality.** The errors are independent of $X$ and independently and identically distributed as $N(0, \sigma^2)$.

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**Theorem 10.5 — Normal Sampling Distributions.** Under TS.1–TS.6,

$$u \mid X \sim N_T(0, \sigma^2 I_T) \quad \Longrightarrow \quad \hat{\beta} \mid X \sim N_{k+1}\left(\beta, \sigma^2 (X'X)^{-1}\right).$$

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Let $\nu = T - k - 1$ and let $R$ have $q$ linearly independent rows. Conditional on $X$,

$$H_0 : \beta_j = \beta_{j,0} \quad \Longrightarrow \quad t_j = \frac{\hat{\beta}_j - \beta_{j,0}}{\mathrm{se}(\hat{\beta}_j)} \sim t_\nu,$$

$$H_0 : R\beta = r \quad \Longrightarrow \quad F = \frac{(R\hat{\beta} - r)'[R(X'X)^{-1}R']^{-1}(R\hat{\beta} - r)}{q\hat{\sigma}^2} \sim F_{q,\nu}.$$

Under these assumptions, the tests and confidence intervals from the cross-sectional inference lecture apply directly.

# Functional Form, Dummy Variables, and Index Numbers

**Functional form.** Logs, quadratics and interactions have the same interpretations as in cross sections. In a log–log distributed lag model, $\delta_0$ is the impact elasticity and $\sum_{j=0}^{q} \delta_j$ is the long-run elasticity.

**Dummy variables.** Indicators can represent seasons, temporary events or changes in regime. Their coefficients compare periods, holding the other regressors fixed.

**Index numbers.** An index expresses a series relative to a base period, often set to 100. Deflating a nominal amount by a price index converts it to purchasing power in base-period prices.

# Trends and Seasonality

# Linear trends and seasonal controls

Many economic time series trend over time; causal interpretation requires accounting for these trends. Ignoring them can create spurious relationships between otherwise unrelated variables.

For quarterly data, consider

$$y_t = \beta_0 + \beta_1 z_t + \eta t + \theta_2 D_{2t} + \theta_3 D_{3t} + \theta_4 D_{4t} + u_t,$$

where $D_{qt} = 1$ in quarter $q$; quarter 1 is the reference category.

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where $D_{qt} = 1$ in quarter $q$; quarter 1 is the reference category.

- $\eta$: holding other factors fixed, the change in $y_t$ per period due to the passage of time.
- $\beta_1$: the change associated with one more unit of $z_t$, controlling for the trend and season.
- $\theta_q$: the difference between quarter $q$ and quarter 1, net of $z_t$ and the trend.

With zero conditional mean and other regressors fixed, expected $y_t$ is linear in $t$.

# Exponential trends and constant growth

Many economic series are better approximated by an **exponential trend**: they grow by a similar percentage, rather than a similar amount, each period.

For $y_t > 0$, model its natural logarithm as a linear trend:

$$\log y_t = \beta_0 + \beta_1 t + u_t, \qquad \mathbb{E}(u_t) = 0.$$

Then $\mathbb{E}(\log y_t) = \beta_0 + \beta_1 t$. Holding the disturbance fixed,

$$\Delta \log y_t = \beta_1, \qquad \frac{y_t - y_{t-1}}{y_{t-1}} = e^{\beta_1} - 1.$$

$\beta_1$ is approximately the per-period growth rate: the trend grows by about $100\beta_1\%$ per period, or exactly $100(e^{\beta_1} - 1)\%$.

For annual data, $\beta_1 = 0.02$ means approximately 2% growth per year.

# Stationarity and Weak Dependence

# Stationary and nonstationary time series

To study relationships between variables from a single time series realization using regression, we need some stability over time.

**Stationary stochastic process.** The stochastic process $\{x_t : t = 1, 2, \ldots\}$ is stationary if for every collection of time indices $1 \leq t_1 < t_2 < \cdots < t_m$, the joint distribution of $(x_{t_1}, \ldots, x_{t_m})$ is the same as the joint distribution of $(x_{t_1+h}, \ldots, x_{t_m+h})$ for all integers $h \geq 1$.

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- The sequence $\{x_t : t = 1, 2, \ldots\}$ is identically distributed.
- $x_t$ and $x_{t+1}$ may be dependent, but their joint distribution must be the same across all time periods.
- Stationarity can be hard to identify from one realization; a mean trend rules it out.

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- Stationarity can be hard to identify from one realization; a mean trend rules it out.

**Covariance stationary process.** A stochastic process $\{x_t : t = 1, 2, \ldots\}$ with finite second moments, $\mathbb{E}(x_t^2) < \infty$, is covariance stationary if

$$\mathbb{E}(x_t) = \mu, \qquad \mathrm{Var}(x_t) = \gamma(0), \qquad \mathrm{Cov}(x_t, x_{t+h}) = \gamma(h).$$

# Weakly dependent time series

A stationary time series process $\{x_t : t = 1, 2, \ldots\}$ is **weakly dependent** if $x_t$ and $x_{t+h}$ are "almost independent" as $h$ increases without bound.

**Examples of weakly dependent time series**

- An independent, identically distributed sequence.
- **Moving average process of order one [MA(1)]:** $x_t = e_t + \alpha_1 e_{t-1}$.
- **Autoregressive process of order one [AR(1)]:** $y_t = \rho_1 y_{t-1} + e_t$, $|\rho_1| < 1$.

Here $\{e_t\}$ is iid with mean zero and finite variance. The AR(1) uses **stationary initialization**: $y_0$ is drawn from its stationary distribution.

# Moving average process of order one [MA(1)]

Let $\{e_t : t = 0, 1, \ldots\}$ be iid, with $\mathbb{E}(e_t) = 0$ and $\mathrm{Var}(e_t) = \sigma_e^2 \in (0, \infty)$. Define

$$x_t = e_t + \alpha_1 e_{t-1}, \qquad t = 1, 2, \ldots.$$

Each observation is a weighted average of the current and preceding innovations.

Adjacent terms share $e_t$, so

$$\mathrm{Cov}(x_t, x_{t+1}) = \alpha_1 \sigma_e^2, \qquad \mathrm{Var}(x_t) = (1 + \alpha_1^2)\sigma_e^2,$$

$$\mathrm{Corr}(x_t, x_{t+1}) = \frac{\alpha_1}{1 + \alpha_1^2}.$$

For example, $\alpha_1 = 0.5$ gives a lag-one correlation of 0.4.

At lags $h \geq 2$, $x_t$ and $x_{t+h}$ share no innovations and are independent. Thus the MA(1) is stationary and weakly dependent.

# Autoregressive process of order one [AR(1)]

Consider

$$y_t = \rho_1 y_{t-1} + e_t, \qquad t = 1, 2, \ldots,$$

where $\{e_t\}$ is iid with mean zero and variance $0 < \sigma_e^2 < \infty$, and is independent of $y_0$.

This is an **AR(1)** process. Its **stability condition** is $|\rho_1| < 1$.

A stable AR(1) is **asymptotically uncorrelated**: observations become uncorrelated as the time between them grows.

Assume **covariance stationarity**, so $\mathbb{E}(y_t) = \mathbb{E}(y_{t-1}) = \mu$. Taking expectations gives

$$\mu = \rho_1 \mu \quad \Longrightarrow \quad (1 - \rho_1)\mu = 0 \quad \Longrightarrow \quad \mu = 0, \qquad \rho_1 \neq 1.$$

The variance $\mathrm{Var}(y_t) = \sigma_y^2$ is also constant over time.

# AR(1): variance and autocorrelation

Since $e_t$ is independent of $y_{t-1}$,

$$\mathrm{Var}(y_t) = \rho_1^2 \mathrm{Var}(y_{t-1}) + \sigma_e^2.$$

Under covariance stationarity, $\mathrm{Var}(y_t) = \mathrm{Var}(y_{t-1}) = \sigma_y^2$, so

$$\sigma_y^2 = \rho_1^2 \sigma_y^2 + \sigma_e^2 \quad \Longrightarrow \quad \boxed{\sigma_y^2 = \frac{\sigma_e^2}{1-\rho_1^2}}.$$

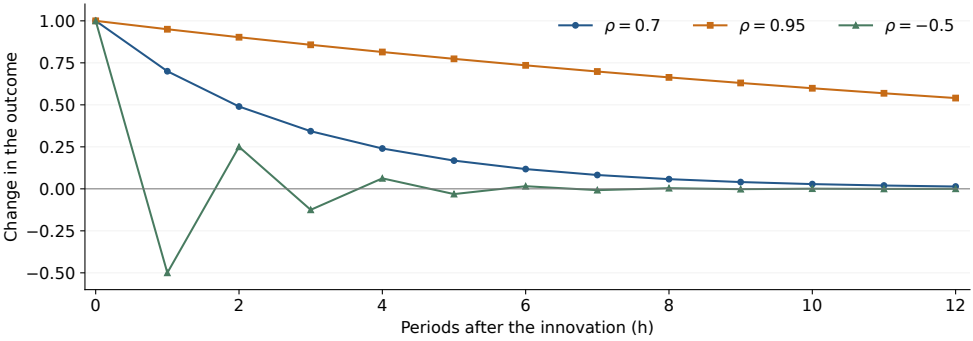
For $h \geq 1$, repeated substitution and independence of future innovations give

$$y_{t+h} = \rho_1^h y_t + \sum_{j=1}^{h} \rho_1^{h-j} e_{t+j} \quad \Longrightarrow \quad \mathrm{Cov}(y_t, y_{t+h}) = \rho_1^h \sigma_y^2.$$

The **autocorrelation function** is therefore

$$r(h) = \mathrm{Corr}(y_t, y_{t+h}) = \frac{\rho_1^h \sigma_y^2}{\sigma_y^2} = \rho_1^h \longrightarrow 0 \quad \text{as } h \to \infty.$$

# AR(1): persistence of a one-unit innovation




Holding later innovations fixed, the path changes by $\rho^h$ after $h$ periods. Negative $\rho$ gives alternating responses. These are model dynamics, not identified policy effects.

# Asymptotic properties of OLS

# Asymptotic assumptions for OLS

**TS.1′ — Linearity and Weak Dependence.** The process $\{(x_t, y_t) : t = 1, 2, \ldots\}$ is stationary and weakly dependent and follows the linear model

$$y_t = \beta_0 + \beta_1 x_{t1} + \cdots + \beta_k x_{tk} + u_t.$$

The required moments and dependence conditions allow LLNs and CLTs for the relevant sample averages.

**TS.2′ — No Perfect Collinearity.**

**TS.3′ — Zero Conditional Mean.** The regressors are contemporaneously exogenous:

$$\mathbb{E}(u_t \mid x_t) = 0, \qquad x_t = (x_{t1}, \ldots, x_{tk})'.$$

Regressors may include lagged outcomes. TS.3′ does not rule out feedback from $y_t$ to future values of an explanatory variable $z$, provided contemporaneous exogeneity holds.

# Consistency of OLS

**Theorem 11.1.** Under TS.1′–TS.3′ and the maintained moment and dependence conditions, the OLS estimators are consistent:

$$\hat{\beta}_j \xrightarrow{p} \beta_j, \qquad j = 0, 1, \ldots, k.$$

- OLS is **consistent, but not necessarily unbiased**: a finite-sample bias can coexist with convergence to the population coefficient.
- We have **weakened exogeneity**: conditioning only on current regressors replaces conditioning on regressors at every date.
- We now require **weak dependence**: a longer history must provide enough new information for sample averages to stabilize.
- Weak dependence will also be crucial for **approximate distributional results** through a time-series CLT.

# OLS in an AR(1) model

Consider the model and conditional-mean restriction

$$y_t = \beta_0 + \beta_1 y_{t-1} + u_t, \qquad \mathbb{E}(u_t \mid y_{t-1}, y_{t-2}, \ldots) = 0.$$

Then

$$\mathbb{E}(y_t \mid y_{t-1}, y_{t-2}, \ldots) = \mathbb{E}(y_t \mid y_{t-1}) = \beta_0 + \beta_1 y_{t-1}.$$

Once $y_{t-1}$ is included, further lags do not change the conditional mean.

Since $x_t = y_{t-1}$, TS.3′ holds. Strict exogeneity fails because $y_t$ is next period's regressor and

$$\mathrm{Cov}(y_t, u_t) = \mathrm{Cov}(\beta_0 + \beta_1 y_{t-1} + u_t, u_t) = \mathrm{Var}(u_t) > 0.$$

Under stability ($|\beta_1| < 1$) and the remaining TS.1′–TS.3′ conditions, OLS is consistent. The slope estimate can still have substantial finite-sample bias, especially near one.

# Asymptotic distribution: assumptions

**TS.4′ — Homoskedasticity.** The errors are contemporaneously homoskedastic:

$$\mathrm{Var}(u_t \mid x_t) = \sigma^2.$$

**TS.5′ — No Serial Correlation.** For $t \neq s$,

$$\mathbb{E}(u_t u_s \mid x_t, x_s) = 0.$$

Serial correlation is often a problem in static and finite distributed lag models: nothing guarantees that their unobservables are uncorrelated over time.

For an AR(1), zero conditional mean given all past outcomes implies no serial correlation in the errors (TS.5′).

# AR(1): absence of serial correlation

In Example 11.3, let $\mathcal{H}_{t-1}$ contain all past outcomes. The model assumes

$$y_t = \beta_0 + \beta_1 y_{t-1} + u_t, \qquad \mathbb{E}(u_t \mid \mathcal{H}_{t-1}) = 0.$$

For $s < t$, the past error can be recovered from past outcomes:

$$u_s = y_s - \beta_0 - \beta_1 y_{s-1} \quad \text{is known given } \mathcal{H}_{t-1}.$$

The regressors $x_t = y_{t-1}$ and $x_s = y_{s-1}$ are also in that history. Iterated expectations therefore give

$$\mathbb{E}(u_t u_s \mid x_t, x_s) = \mathbb{E}\left[ u_s \underbrace{\mathbb{E}(u_t \mid \mathcal{H}_{t-1})}_{0} \mid x_t, x_s \right] = 0.$$

Thus TS.5′ follows from the correctly specified conditional mean; merely adding a lag does not guarantee it.

# Asymptotic distribution of OLS

**Theorem 11.2.** Under TS.1′–TS.5′ and the maintained moment and CLT conditions, OLS is asymptotically normally distributed:

$$\sqrt{T}(\hat{\beta} - \beta) \xrightarrow{d} N(0, \sigma^2 Q^{-1}), \qquad Q = \operatorname{plim}_{T\to\infty} \frac{X'X}{T} \text{ nonsingular.}$$

The usual OLS standard errors and the associated $t$, $F$ and LM tests are asymptotically valid.

For the coefficient and joint-restriction tests introduced earlier, under their respective null hypotheses,

$$t_j \xrightarrow{d} N(0,1), \qquad qF \xrightarrow{d} \chi^2_q,$$

where $q$ is the number of independent restrictions.

# Example 11.4: Efficient Markets Hypothesis

Can past stock returns help predict the current return? Let $y_t$ be the weekly percentage return on the NYSE composite index.

The version of the **Efficient Markets Hypothesis (EMH)** considered here says that past returns do not change the expected return:

$$\mathbb{E}(y_t \mid y_{t-1}, y_{t-2}, \ldots) = \mathbb{E}(y_t).$$

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$$\mathbb{E}(y_t \mid y_{t-1}, y_{t-2}, \ldots) = \mathbb{E}(y_t).$$

Use an AR(1) as the alternative model:

$$y_t = \beta_0 + \beta_1 y_{t-1} + u_t, \qquad H_0 : \beta_1 = 0.$$

Under the null, $\mathbb{E}(u_t \mid y_{t-1}) = 0$ and the conditional-mean restriction rules out serially correlated errors.

With homoskedasticity and the required weak dependence and moments, Theorem 11.2 justifies the usual $t$ test.

# Example 11.4: evidence from weekly returns

For January 1976–March 1989,

$$\widehat{return}_t = \underset{(0.081)}{0.180} + \underset{(0.038)}{0.059}\, return_{t-1}, \qquad n = 689, \quad R^2 = 0.0035.$$

# Example 11.4: evidence from weekly returns

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Parentheses contain standard errors. The slope statistic is $t \approx 1.55$: we do not reject $H_0 : \beta_1 = 0$, even at 10%.

A one-lag model can miss predictability at longer lags. Adding $y_{t-2}$ gives the joint test

$$H_0 : \beta_1 = \beta_2 = 0, \qquad F \approx 1.65, \quad p \approx 0.193 \quad (n = 688).$$

The two-lag test also fails to reject. These results find little evidence of return predictability from these lags; they do not establish the hypothesis.

# Highly persistent time series in regression analysis

# Highly persistent series: a random walk

Set $\rho_1 = 1$ in the AR(1), with iid innovations satisfying $\mathbb{E}(e_t) = 0$, $\mathrm{Var}(e_t) = \sigma_e^2 > 0$, and $y_0$ independent of $e_t$ for all $t \geq 1$:

$$y_t = \alpha_0 + y_{t-1} + e_t.$$

Here $\alpha_0$ is the **drift**. Take $\alpha_0 = 0$ (**no drift**), and $y_0 = 0$ (a fixed initial value). Then

$$y_t = \sum_{s=1}^{t} e_s, \qquad \mathbb{E}(y_t) = 0, \qquad \mathrm{Var}(y_t) = t\sigma_e^2.$$

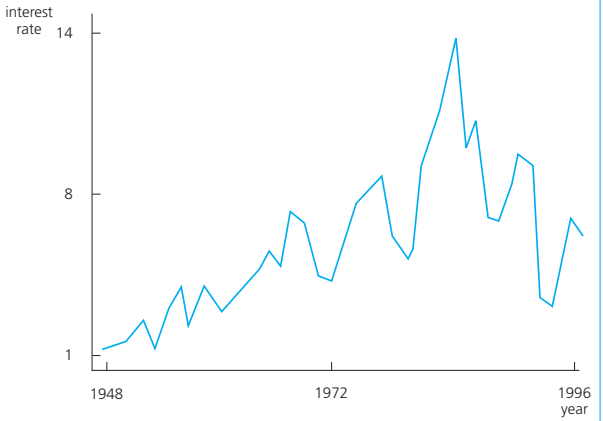
Each innovation permanently shifts the path. For $t \geq 1$ and $h \geq 1$,

$$\mathbb{E}(y_{t+h} \mid y_t) = y_t, \qquad \mathrm{Corr}(y_t, y_{t+h}) = \sqrt{\frac{t}{t+h}}.$$

The random walk is **not covariance stationary**: its variance and its correlation at a fixed lag depend on the calendar date $t$.

# An example: the three-month Treasury-bill rate

**FIGURE 11.2** The U.S. three-month T-bill rate, for the years 1948–1996.




The US three-month T-bill rate is an example of a series often modeled as a **random walk**.

Persistent movements need not follow a deterministic upward or downward trend.

A time-series plot alone cannot establish a unit root.

Source: Wooldridge (5th ed.), Figure 11.2, p. 394. Rates are in percent.

# Transformations on highly persistent time series

A weakly dependent process is **integrated of order zero**, or $I(0)$. A unit-root process is **integrated of order one**, or $I(1)$: its level is not weakly dependent, but its first difference is $I(0)$.

**Random walk.** Differencing removes the accumulated innovations:

$$y_t = \alpha_0 + y_{t-1} + e_t \quad \Longrightarrow \quad \Delta y_t = y_t - y_{t-1} = \alpha_0 + e_t.$$

With iid $e_t$, the differences are iid; with weakly dependent increments, they are weakly dependent. No drift means $\alpha_0 = 0$.

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With iid $e_t$, the differences are iid; with weakly dependent increments, they are weakly dependent. No drift means $\alpha_0 = 0$.

**Deterministic linear trend.** Differencing also removes a linear mean trend:

$$y_t = a + bt + v_t \quad \Longrightarrow \quad \Delta y_t = b + v_t - v_{t-1}.$$

If $v_t$ is mean-zero and $I(0)$, then $\mathbb{E}(\Delta y_t) = b$ is constant. This is a **trend-stationary** case: the deviations $v_t$ are already $I(0)$.

# Time series: what we covered and what comes next

**Covered today**

- Temporal ordering; static and distributed lag models; functional form, trends and seasonality.
- Finite-sample and asymptotic OLS: exogeneity, dependence, estimation and inference.
- Stationarity, MA(1) and AR(1); persistence, random walks, $I(0)/I(1)$ and differencing.

**Important topics beyond today**

- Diagnosing serial correlation and adjusting inference: tests, HAC standard errors and GLS.
- Unit-root tests, spurious regression and cointegration.
- Estimating forecasts and evaluating their accuracy; multivariate and volatility models.